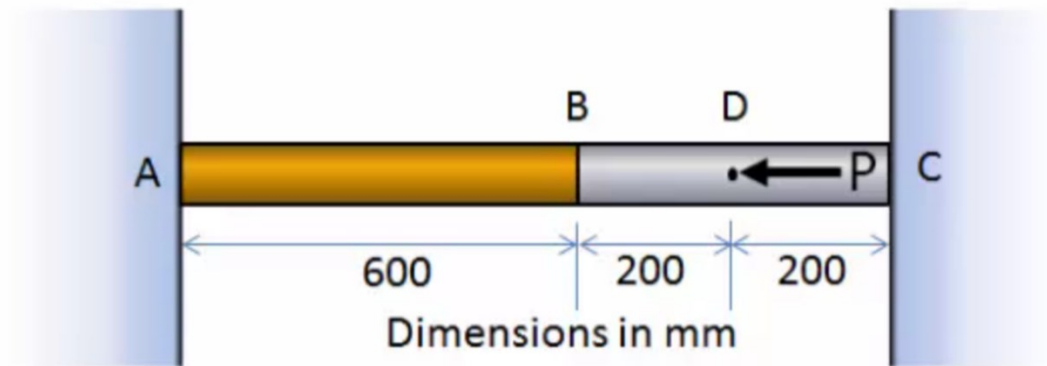


Big Extra Exercise for Solid Mechanics (EMCH260, Fall 2021)

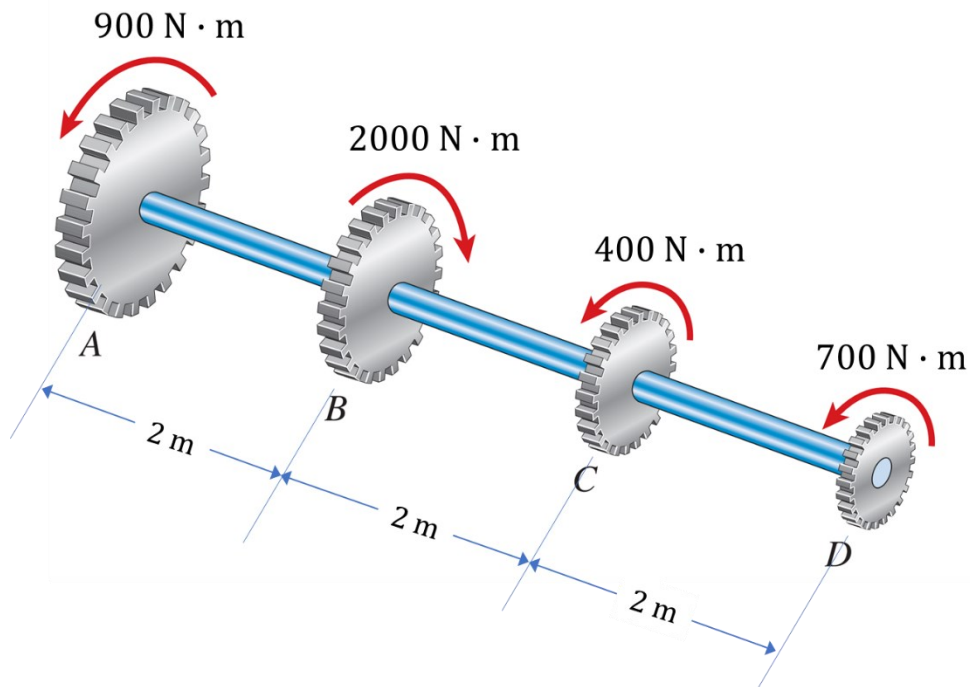
NOTE: For the big extra exercise, you'll earn maximum 6 **net** points that will be added on top of your final grade (in 100-point scale). For example, if you get 80 for the big extra exercise submission, you'll earn $80/100 \times 6 = 4.8$ net point to be added on your final grade (net value!).

Submission due: 8:30 AM, Friday, Dec. 3rd 2021

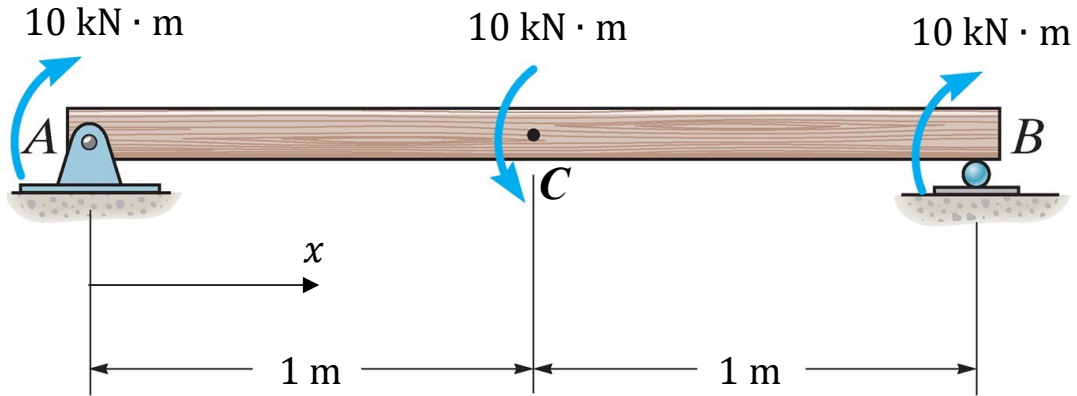
1. The bar has constant diameter of 100 mm and is composed of two different materials. The two ends A and C are fixed to the wall. An external load P of 380 kN is applied at point D . The modulus of elasticity of part AB and BC is 80 GPa and 200 GPa, respectively. Determine the reactions at ends A and C . (10' in total)



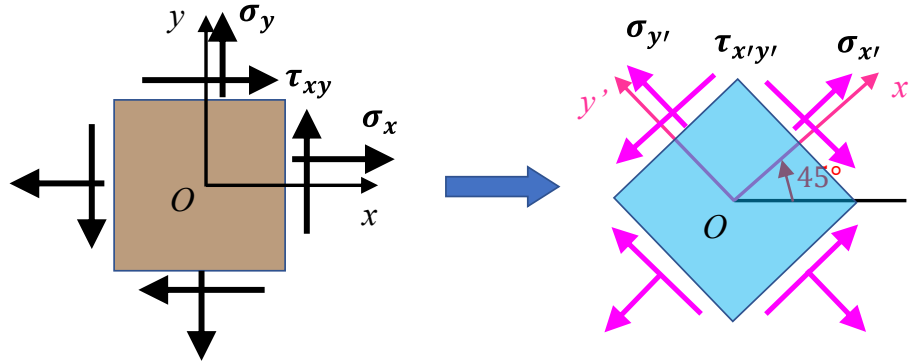
2. The allowable shear stress in the shaft is 70 MPa and the allowable angle of twist of end A relative to end D is 0.01 rad. What is the required minimum diameter d of the shaft if it has a solid cross section? The shear modulus of elasticity of the shaft is 80 GPa. (15' in total)



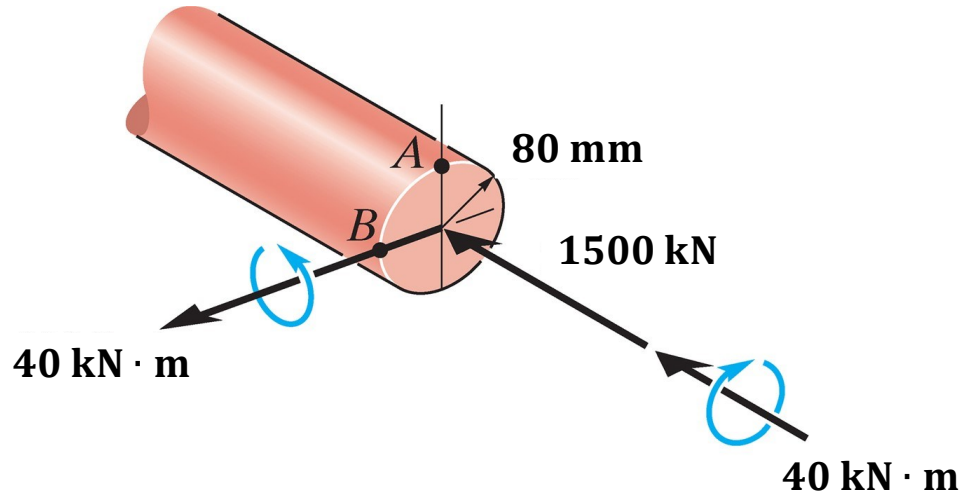
3. Determine the absolute maximum normal stress in the beam with external loading shown below. The beam has a uniform square cross-section with lateral size $a = 0.1$ m. Note: the shear and moment diagrams can be calculated either by section method or graphical method. (15' in total)



4. For a plane stress state, in the original Oxy system we know the component $\sigma_x = 120$ MPa and $\tau_{xy} = 60$ MPa. After the element (see the left figure) is rotated counterclockwise by 45° , we know the stress component $\tau_{x'y'} = -60$ MPa in the new $Ox'y'$ system (see right figure). Determine the two principal stresses and the absolute maximum shear stress ($\tau_{\max}^{\text{abs}}$). (10' in total)

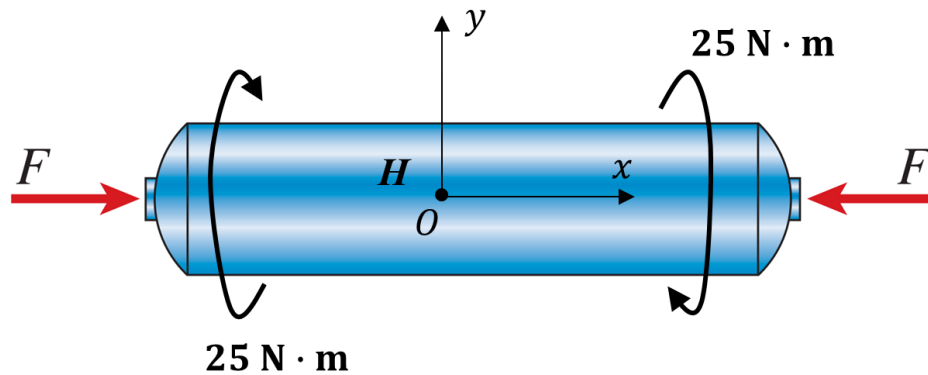


5. The internal loadings at a cross section through the shaft consist of an axial force of 1500 kN, a bending moment of 40 kN · m, and a torsional moment of 40 kN · m. The radius of the shaft is 80 mm. Determine the principal stresses at the surface point *A* and the absolute maximum shear stress at this point. (15' in total)



6. A thin-walled cylindrical pressure vessel of inner radius $r = 0.6$ cm and a thickness of 0.06 cm is subjected simultaneously to internal gas pressure $p = 3.5$ MPa, a compressive force $F = 800$ N acting at the ends, and a torsional load of $25 \text{ N} \cdot \text{m}$ (see figure). Determine the following quantities (**15'** in total):
- (a) the principal stresses at a point H on the surface of the vessel and the absolute maximum shear stress; (**10'**)
 - (b) the normal strain along x -axis (ϵ_x) and the shear strain (γ_{xy}) at the same point H . See the established Oxy coordinate system. The Young's modulus of the material is 200 GPa and the Poisson's ratio is 0.3. (**5'**)

Hint: the cross-section of the vessel can be regarded as a hollow tube and the cross-section area can be calculated as $\pi(0.0066^2 - 0.006^2)$. You can do the same treatment for the polar moment inertia.



7. The compound shaft is composed of bronze cylindrical segment AB and steel cylindrical segment BC . The two ends of the compound shaft are fixed to rigid supports. The external torque (T) is applied at B . The shear modulus of segment AB and BC is 40 GPa and 80 GPa, respectively. The length of the segment AB and BC is 2 m and 1 m, respectively. The diameter of the segment AB and BC is 0.05 m and 0.025 m, respectively. Determine the external torque T applied to point B so that one of the principal stresses for a point H on the surface of the segment AB is -100 MPa. (20' in total)

